Chronology Protection?

In February 1989, as the hoopla in the press was beginning to subside, and while Echeverria, Klinkhammer, and I were struggling with Polchinski's paradox, I flew to Bozeman, Montana, to give a lecture. There I ran into Bill Hiscock, a former student of Charles Misner's. As I have with so many colleagues, I pressed Hiscock for his views on wormholes and time machines. I was searching for cogent criticisms, new ideas, new viewpoints.

"Maybe you should study electromagnetic vacuum fluctuations," Hiscock told me. "Maybe they will destroy the wormhole when infinitely advanced beings try to turn it into a time machine." Hiscock had in mind the thought experiment in which my wife Carolee (assumed to be infinitely advanced) is flying back to Earth in the family spacecraft with one wormhole mouth, while I sit on Earth with the other mouth, and the wormhole is on the verge of becoming a time machine (Figures 14.7 and 14.8 above). Hiscock was speculating that electromagnetic vacuum fluctuations might circulate through the wormhole in the same manner as did bits of radiation in Figure 14.8; and, piling up on themselves, the fluctuations might become infinitely violent and destroy the wormhole.

I was skeptical. A year earlier, on my drive home from Chicago, I had convinced myself that bits of radiation, circulating through the wormhole, will *not* pile up on themselves, create an infinitely energetic beam, and destroy the wormhole. By defocusing the radiation, the wormhole saves itself. Surely, I thought, the wormhole will also defocus a circulating beam of electromagnetic vacuum fluctuations and thereby save itself.

On the other hand, I thought to myself, time machines are such a radical concept in physics that we must investigate anything which has any chance at all of destroying them. So, despite my skepticism, I set out with a postdoc in my group, Sung-Won Kim, to compute the behavior of circulating vacuum fluctuations.

Though we were helped greatly by mathematical tools and ideas that Hiscock and Deborah Konkowski had developed a few years earlier, Kim and I were hampered by our own ineptness. Neither of us was an expert on the laws that govern the circulating vacuum fluctuations: the laws of quantum fields in curved spacetime (Chapter 13). Finally, however, in